Sales Analysis Dashboard – Project Documentation

Author: Anisha Kumari
Tool: Microsoft Excel **Project type:** Data Analytics / Business Intelligence Dashboard

1. Project Overview

This project is an interactive **Sales Analysis Dashboard** built entirely in Microsoft Excel. It turns raw order data into a single-page, business-ready view of revenue performance, so a stakeholder can see how much the business earns, when and where it earns it, and which products and occasions drive sales.

The dashboard uses PivotTables and PivotCharts connected to slicers and timeline filters, so every chart responds instantly when the viewer filters by date or occasion.

2. Objectives

- Track overall business performance through headline KPIs.
- Understand revenue trends across months, hours of the day, and days of the week.
- Identify the top-performing products, cities, and gifting occasions.
- Give users an interactive way to explore the data without touching the underlying tables.

3. Tools & Features Used

Tool / Feature	How it was used
Microsoft Excel	Core platform for data preparation, analysis, and dashboard design
PivotTables	Summarised revenue by month, hour, product, city, occasion, and day
PivotCharts	Visualised each PivotTable as line and column charts
Slicers	Interactive filter for **Occasion** (Anniversary, Birthday, Diwali, Holi, Raksha Bandhan, Valentine's Day)
Timeline Filters	Date filters on **Order_Date** and **Delivery_Date** with month-level selection
Excel Formulas & Functions	Derived fields (such as month, hour, and weekday) and KPI calculations
KPI Cards	Highlight Total Revenue and Average Customer Spending at the top of the dashboard
Dashboard Visualization	Consistent layout, colour theme, and titled chart panels on one screen

4. Key Metrics

- **Total Revenue:** $3,520,984.00
- **Average Customer Spending:** $3,520.98 (implies roughly 1,000 orders in the dataset)

- **Monthly Revenue**
- **Top 5 Products**
- **Top 10 Cities**
- **Occasion-wise Revenue**
- **Day-wise Sales Analysis**

5. Dashboard Layout

The dashboard is organized in three zones:

1. **Header and KPI row:** a branded title banner ("Sales Analysis") followed by two KPI cards for Total Revenue and Average Customer Spending.
2. **Chart grid (2 rows x 3 charts):**
    - *Revenue in Months* – line chart of monthly revenue trend, January to December.
    - *Revenue by Hours* – line chart showing revenue across hours of the day (0–23).
    - *Top 5 Revenue Products* – column chart of the highest-earning product categories (Colors, Plants, Raksha Bandhan, Soft Toys, Sweets).
    - *Revenue by Top 10 Cities* – column chart comparing the leading cities (Bhatpara, Bidhannagar, Bilaspur, Dhanbad, Dibrugarh, Guntakal, Haridwar, Imphal, Kavali, North Dumdum).
    - *Revenue by Occasions* – column chart comparing revenue across gifting occasions.
    - *Revenue by Days* – column chart of revenue by day of the week, Sunday to Saturday.
3. **Filter panel (right side):** Delivery Date timeline, Order Date timeline, and the Occasion slicer.

6. Methodology

1. **Data preparation:** Cleaned the order dataset and used Excel formulas to extract month, hour, and weekday from the order and delivery timestamps.
2. **Analysis with PivotTables:** Built a separate PivotTable for each metric (month, hour, product, city, occasion, day).
3. **Visualization:** Converted each PivotTable into a PivotChart and chose the chart type to match the question: lines for trends over time, columns for category comparisons.
4. **KPI calculation:** Calculated total revenue and average spending per customer and displayed them in KPI cards.
5. **Interactivity:** Connected the slicers and timelines to all PivotTables through Report Connections, so one selection filters the whole dashboard.
6. **Formatting:** Applied a consistent green palette, chart titles, and a clean grid layout for readability.

7. Key Insights

These observations come from reading the dashboard and should be re-checked against the live pivots:

- **Revenue is uneven across the year.** The monthly trend shows clear peaks around February and August, with much lower revenue in most other months.
- **Sweets lead the top 5 products** by a wide margin, well ahead of Colors, Plants, Raksha Bandhan items, and Soft Toys.
- **Revenue by hour fluctuates through the day**, rising from early hours to several peaks later in the day, which can guide timing for promotions.
- **Occasion-driven demand:** Anniversary, Holi, and Raksha Bandhan stand out among the occasions, and Tuesday is the strongest day of the week.
- **City performance is fairly close** among the top 10, with Imphal, Dhanbad, and Kavali near the top of the chart.

8. Skills Demonstrated

- Data Analysis
- Data Visualization
- KPI Reporting
- Business Intelligence
- Interactive dashboard design in Excel

9. Possible Future Improvements

- Add a profit or margin view alongside revenue.
- Add a year-over-year or month-over-month growth indicator to the KPI cards.
- Replace the default "Axis Title" labels with meaningful axis names and units.
- Move the same model to Power BI or Looker Studio for web sharing.

10. Short Summary

Designed an interactive Excel Sales Analysis dashboard using PivotTables, PivotCharts, slicers, and timeline filters. Tracked Total Revenue ($3.52M) and Average Customer Spending ($3,520.98), and analysed monthly, hourly, city-wise, product-wise, occasion-wise, and day-wise sales to surface revenue trends and top performers.